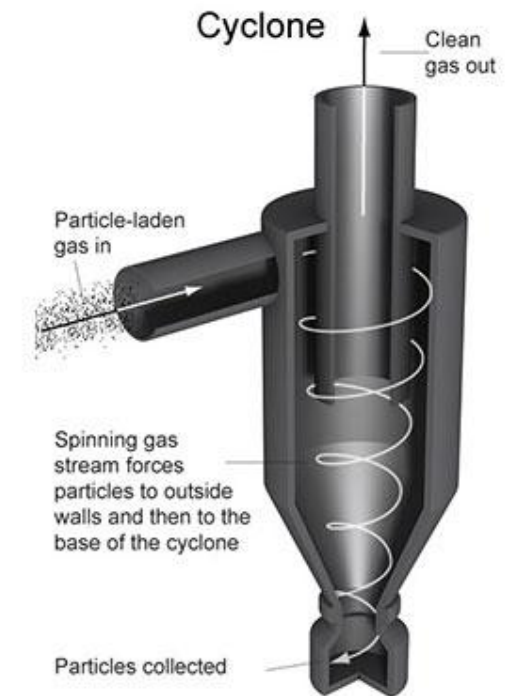


- Sound = pressure wave

Filters

- Coffee filter
- Particle filter (to measure pollutions)
- Filters in Signal Processing (remove or attenuate some frequencies while keeping the others without any change)

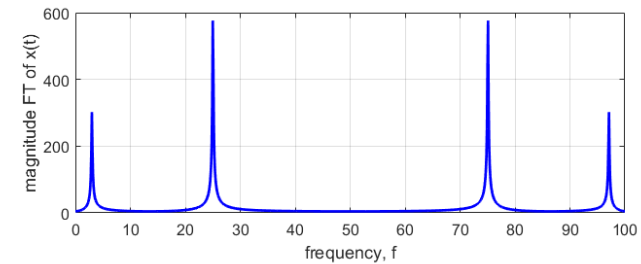
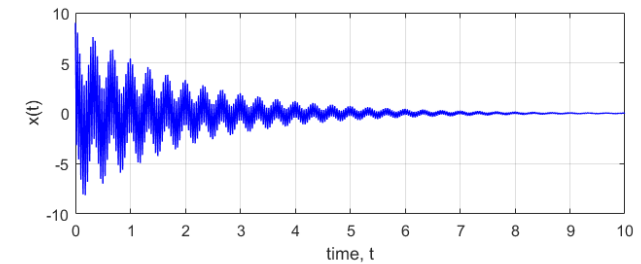
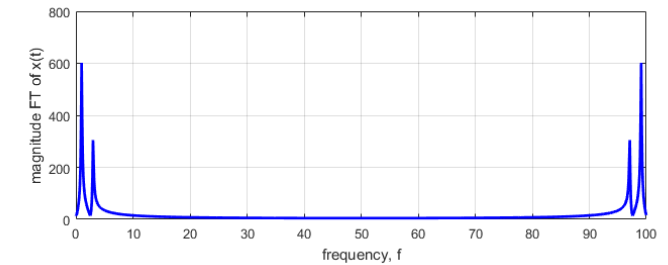
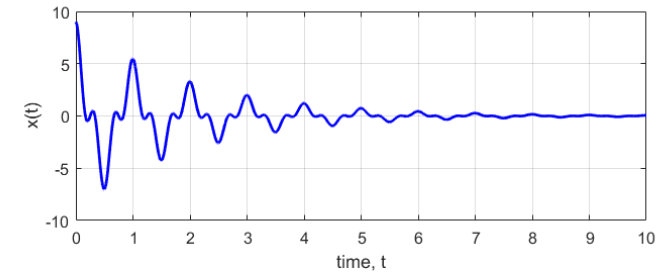


Recall (Lecture 3)

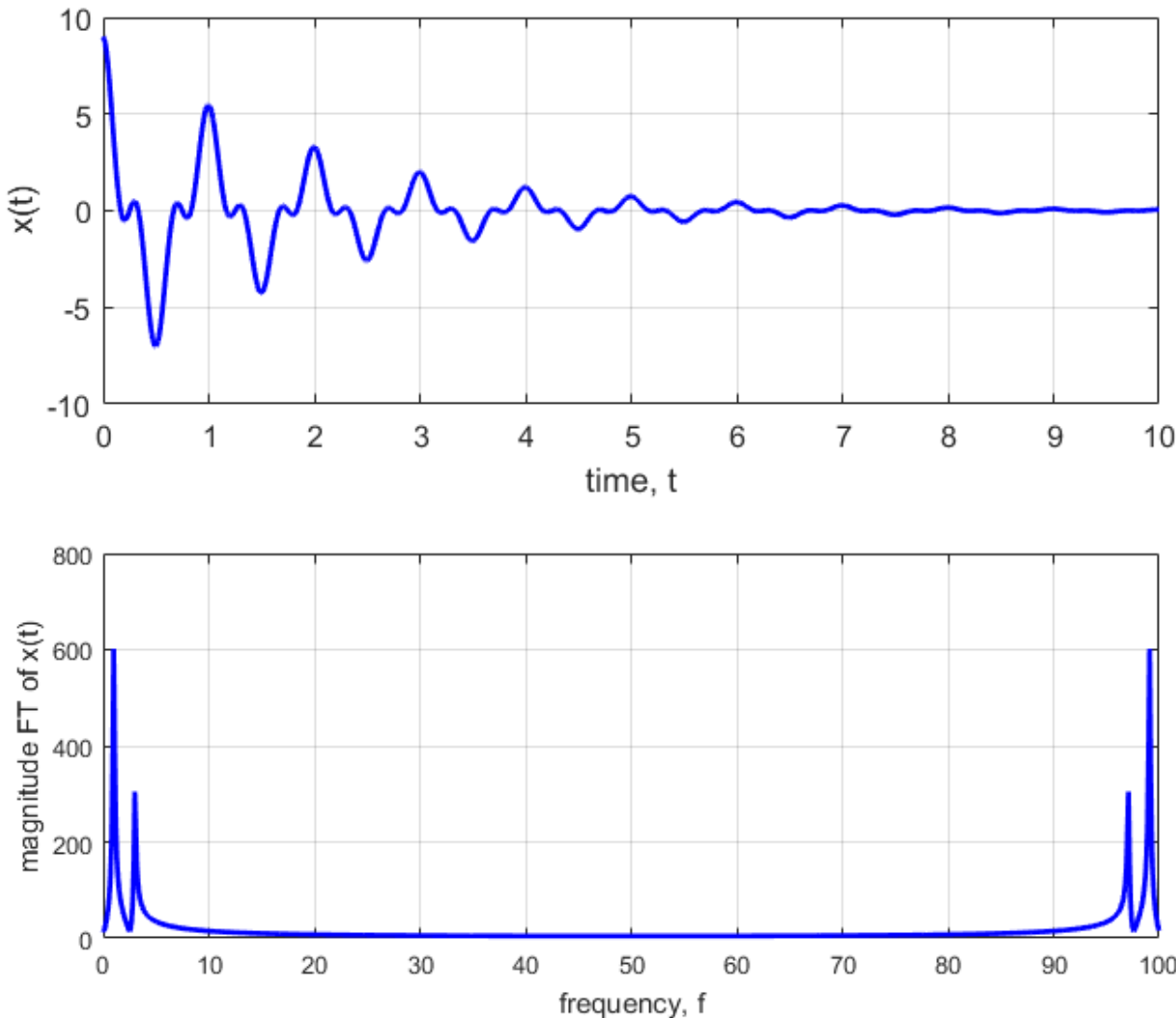
A signal can be presented as a mixture of sines and cosines (complex exponentials)

$$\int_{-\infty}^{+\infty} X(f) e^{-j2\pi f t} df = x(t)$$

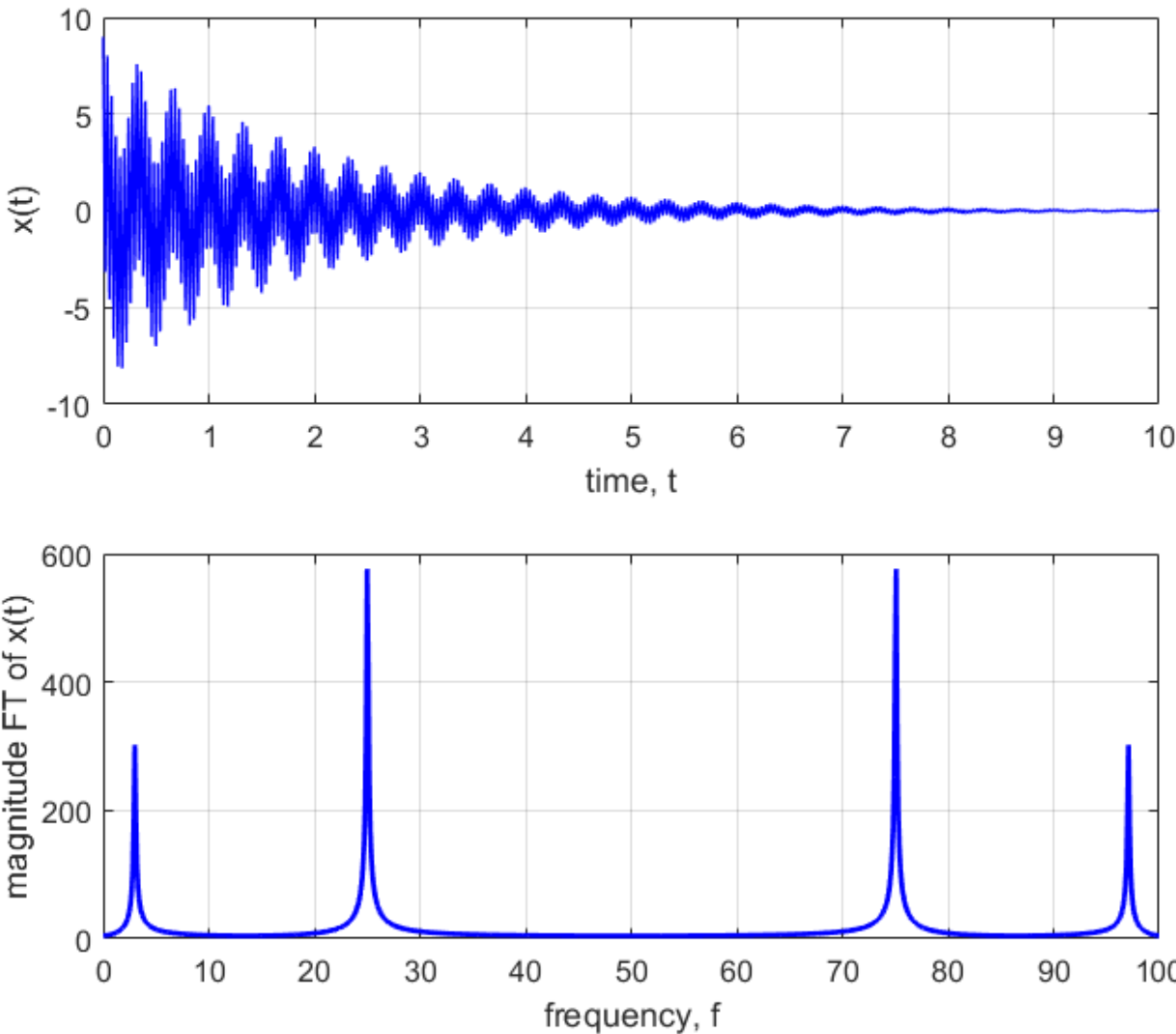
$X(f)$ tell us how much each frequency contributes



Illustration

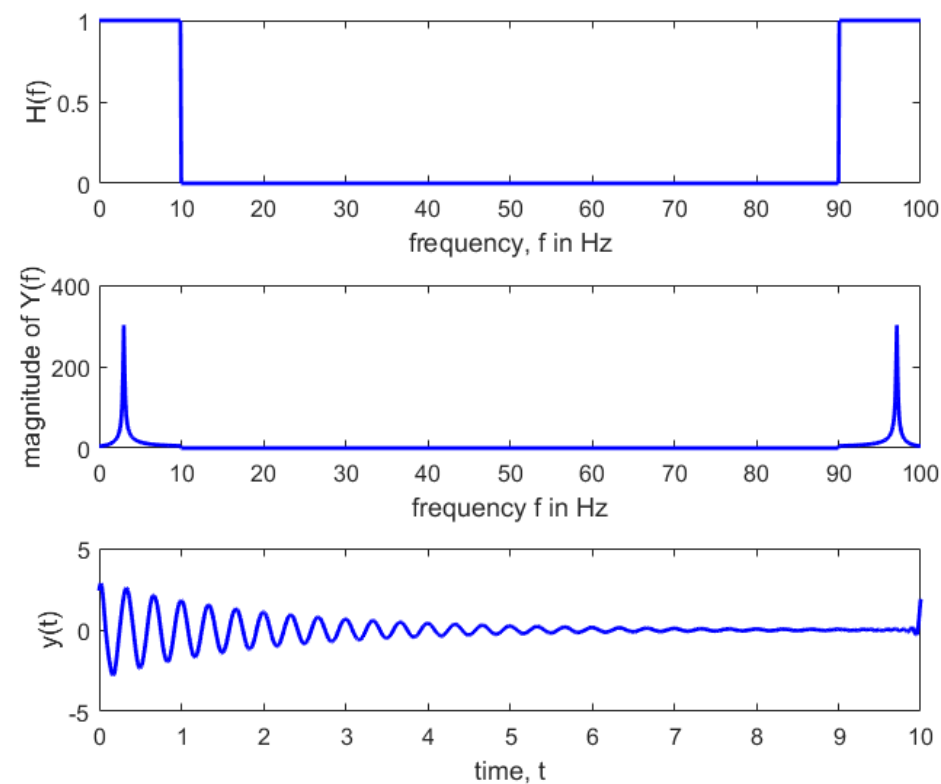


Illustration



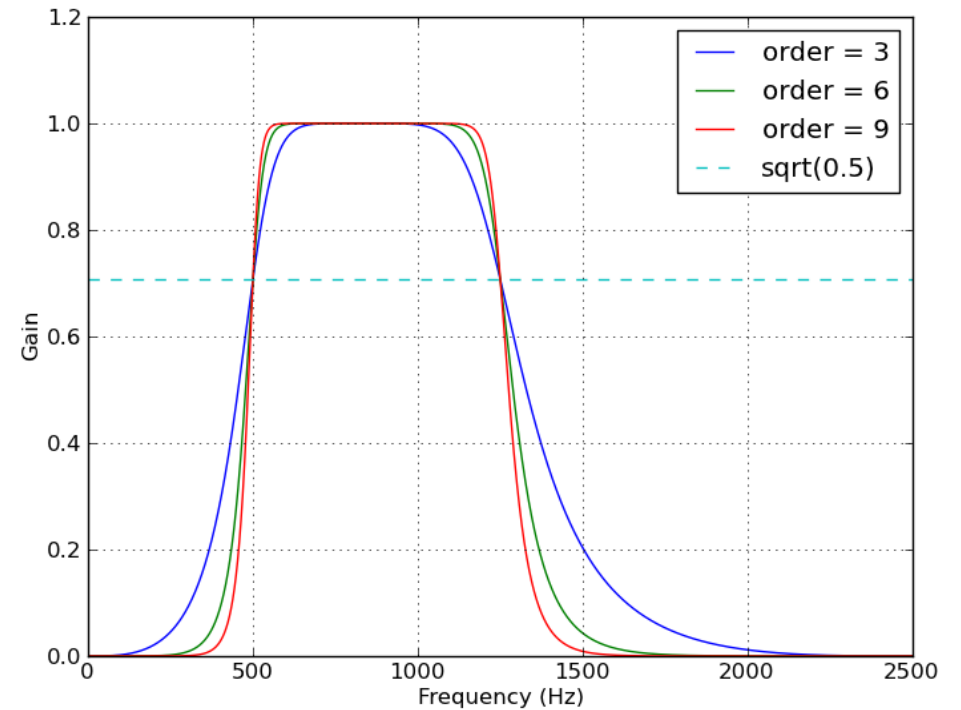
How are filters applied?

- $x(t) \xleftrightarrow{FT} X(f)$
- Filter response $h(t)$ to a narrow pulse on the input (think of it as a delta-function)
- $h(t) \xleftrightarrow{FT} H(f)$
- The response of filter on a composed input $x(t)$ is a correlation of $x(t)$ and $h(-t)$
- By Correlation Theorem:
- $Y(f) = X(f)H(f)$
- $y(t)$ is the output of the filter.

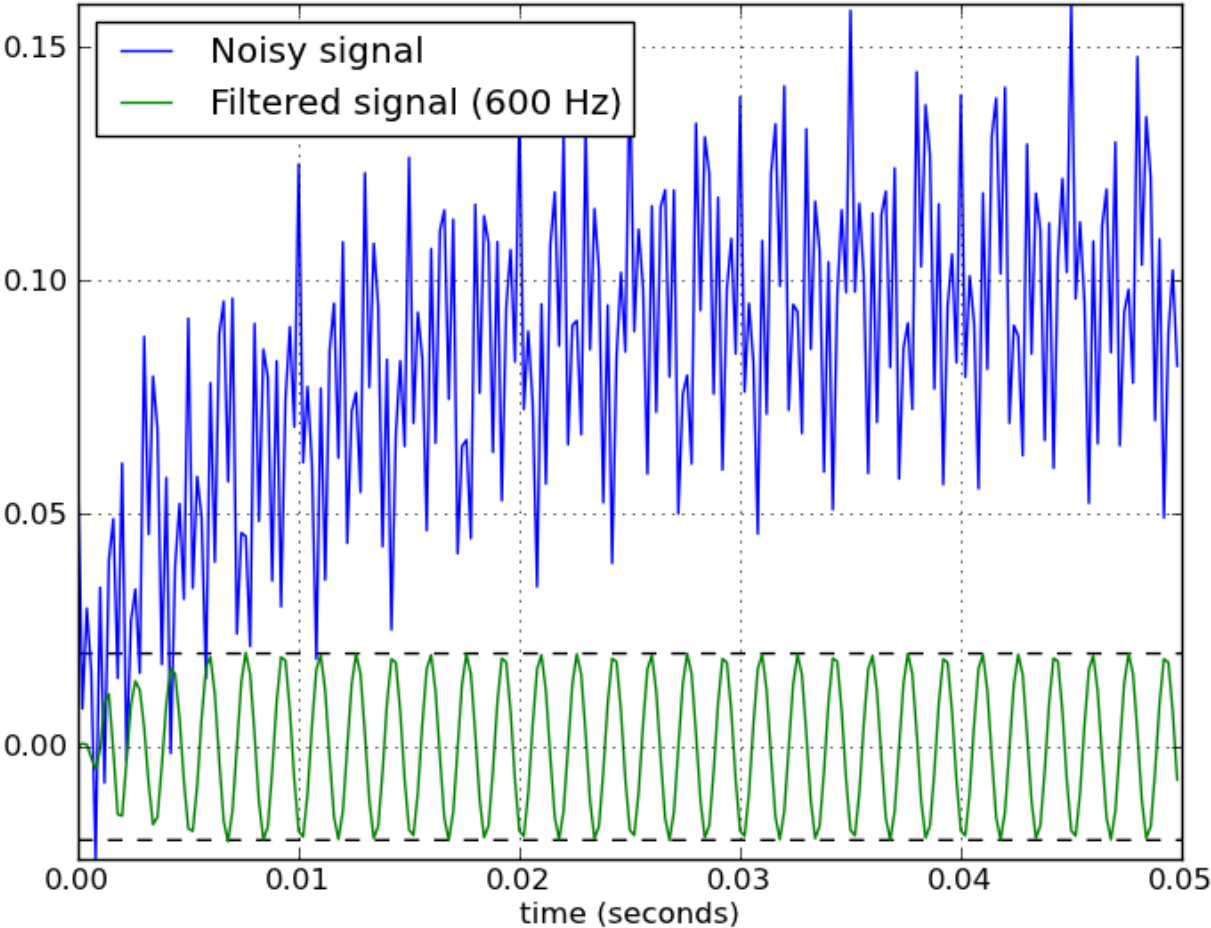


Four types of filters

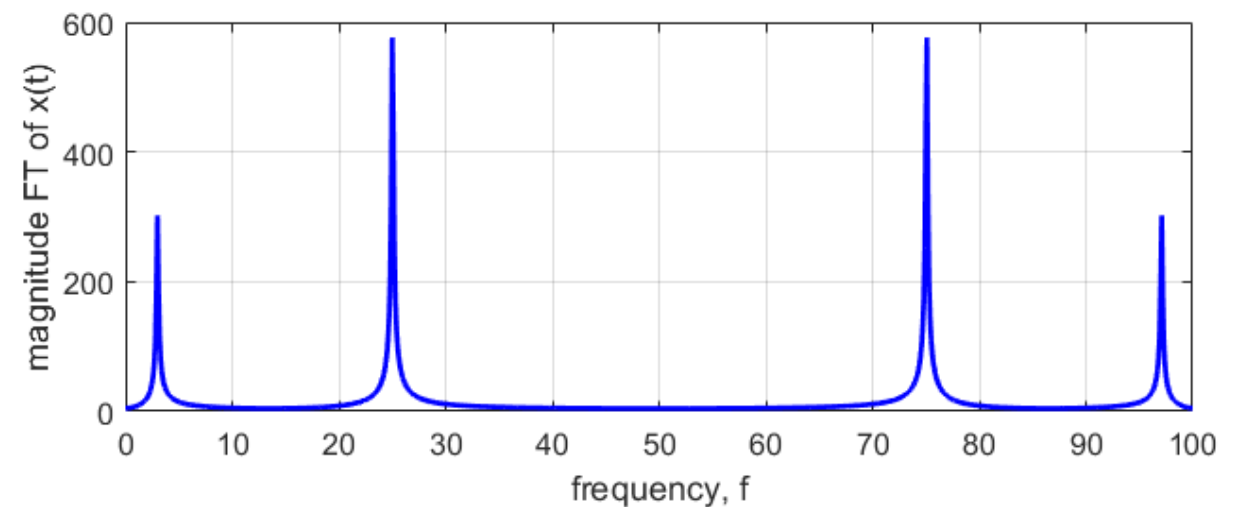
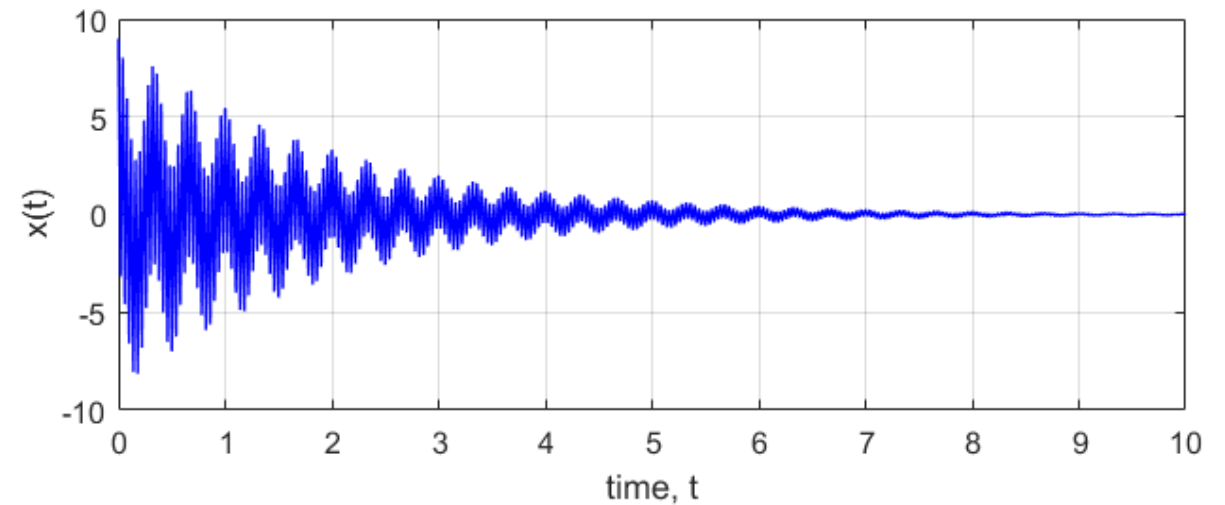
- Low pass = suppresses high frequencies, passes low frequencies
- High pass = passes high frequencies and suppresses low
- Band pass = passes medium frequencies, suppresses low and high
- Band stop



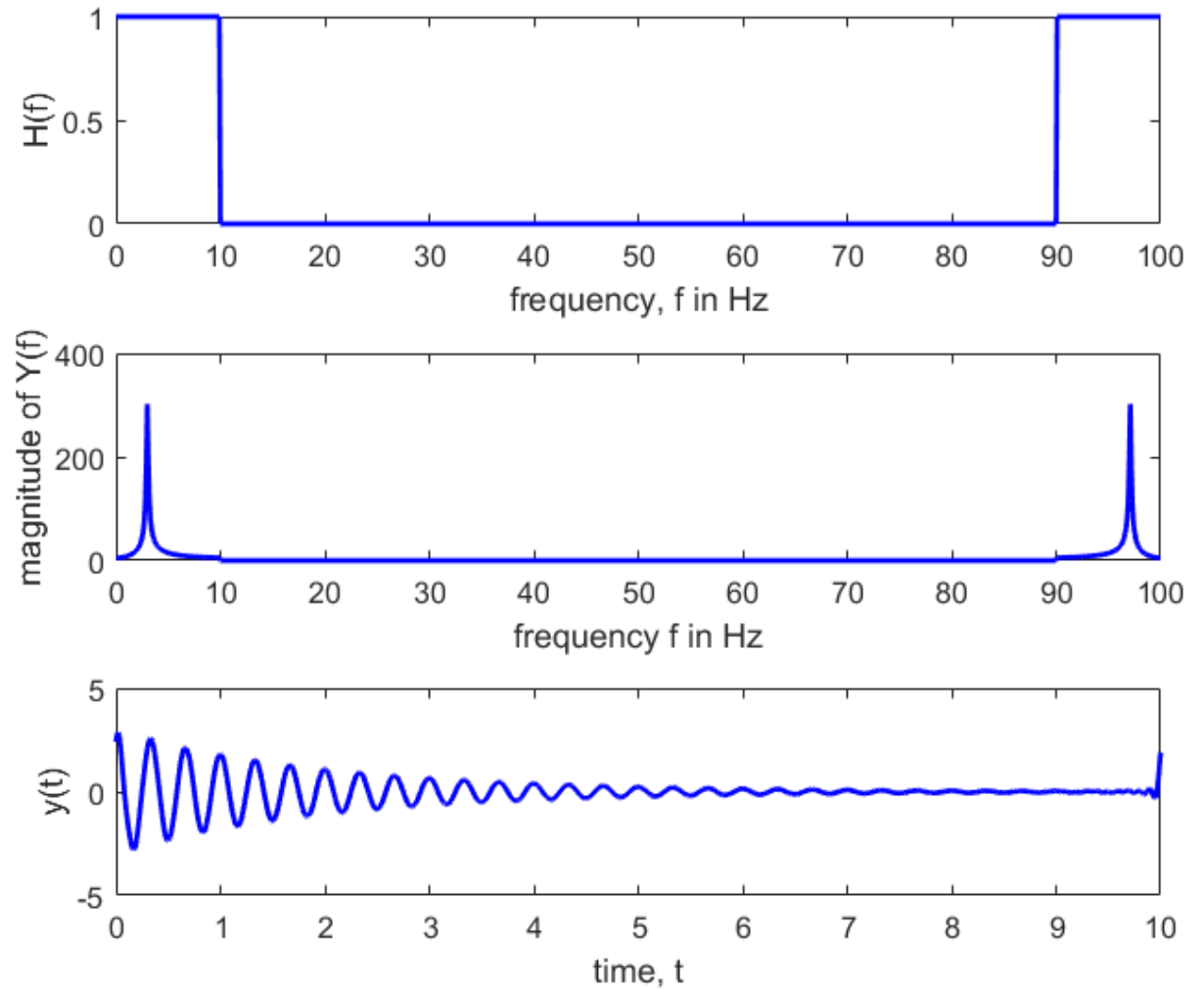
Illustration



Low pass filtering: the original signal and its FT



Signal and its FT after low pass filtering



Filtering in 2D

- Simulated pulse in filter-bank format (image)
- $DM = 375$
- $S/N = 25$
- Pulse width $T = 50$ ms
- Apply low-pass filter in 2d FT space

